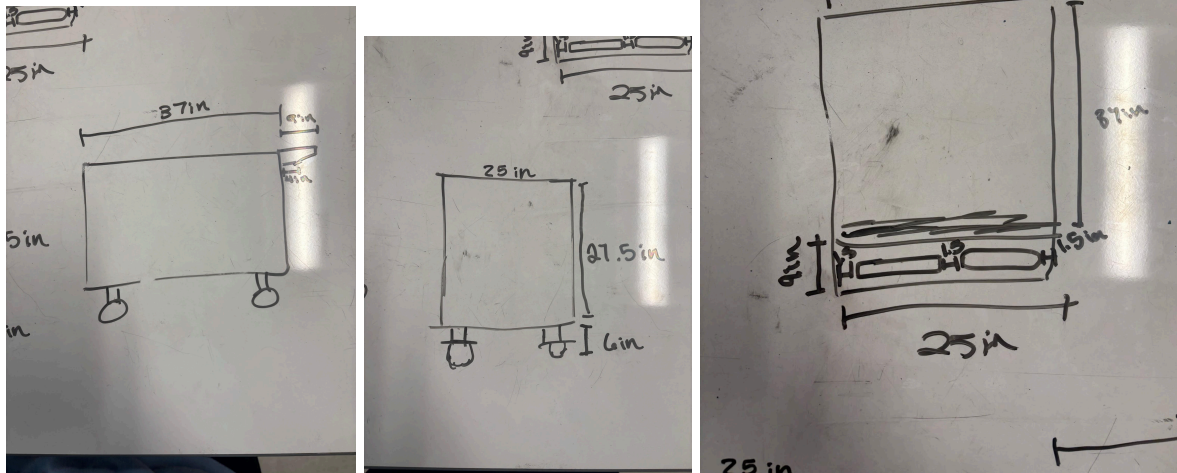


Element E

During prototype creation, our team used various parts of the STEM principles to try and create a successful solution. So far, our prototype has only been created using wood, pvc pipes, clamps, and many other technical tools. Initially, We mainly used the mathematics portion of the STEM principles. Prior to prototype creation, we took measurements of the entire cart in case we needed them.



In the end, we decided to measure the length from the bottom of the cart to where the rods pass through the original cart handle. This was so we could determine the spot to carve our wood a little so we were able to fit the wood plank through the handle and into the bottom section. The picture on the right below depicts Nic whittling down part of the

wood so we can add a rubber lacing around that area. This area slides between the original cart's handle so our design doesn't move, but is also very flexible and won't break.



We also had to measure the thickness of the PVC pipe rods that we were going to use. This was because we needed to drill the correct size hole into our wood planks as Nic is doing down in the picture below. While we were working on drilling the hole, part of the wood plank broke off. At first, we thought of this as a setback. But then we realized that this new form of wood might be able to clip onto the bottom. This was a different solution

that we had tried at first, but so far it worked well, so we also tried to develop this project too. This also demonstrates a strong foundation in the engineering principles as we were able to put a plan into place, but still keep improving our design through setbacks.



During our time prototyping, we had the wonderful opportunity to meet with Vanderbilt teachers and students in different engineering departments. We were able to meet with two of them and they each gave us some feedback. First, we met with Dr. London, an associate professor of mechanical engineering and an assistant provost of academic opportunity and belonging. She also had a background in human factors and a little in biomed. She had some great ideas for iteration that we have taken into consideration. After we let her try our design, she mentioned that it would be helpful if there were instructions on the side so that people using the cart would know how to take the handle out. Our team hadn't thought about this, but we will incorporate it into our final design. Secondly, she helped us tackle the universal cart criteria. She insisted that we insert a longer PVC pipe with clamps on the bottom of the cart so that carts with wider and smaller bases alike would still be able to use this design. This image below shows the two pipes at the top, and one more at the bottom, which we added to make the design more secure. Now all we needed to do was cut the pipe down so that it can fit through doors and narrow spaces.



We also discussed iterations with a Vanderbilt student named Jared, who is in medical engineering with a basis in biomed as well. Though we didn't talk to Jared for that long, he was able to give us one piece of advice that has well developed our prototype as a whole. He thought that a locking mechanism for the PVC pipe that inserts through the two holes in our wood rods would be useful so our handle didn't shift so easily. We did end up including something of the sort into our design. Instead of a locking mechanism, we added a rubber layer onto the PVC pipe where it sticks into the wooded planks and is harder to move. This allows for the handle to stay upright even without anyone using it. Below, you can see the red rubber and hot glue that adds friction so the handle doesn't move so easily.



Both of these feedback and iteration shows a basis in the STEM principles as we somewhat followed the Engineering Design Process to iterate and test our products as we made changes. As we made changes, we let Mr. Kiser test our design and describe how it felt so we could continue to further improve our idea.